Growth of ζ and of Dirichlet L-functions to the right of 7/8: explicit consequences of the uniform zero-free half-plane

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October 27, 2026 — draft v3 (density routes checked with the ANTEDB engine), not yet externally reviewed

Abstract

The uniform zero-free half-plane $\operatorname{Re} s > 7/8$ for all Dirichlet L-functions (OpenAI, September 2026, Lean-verified) implies, by the classical Borel–Carathéodory/three-circles argument, that $L(\sigma+it,\chi)^{\pm1} \ll_{\delta,\varepsilon} (q(|t|+3))^{\varepsilon}$ uniformly in q and t for $\sigma \ge 7/8+\delta$. This is the Dirichlet analogue of Lemma 4.9 ("global assertion") of the OpenAI manuscript, which is stated there, as an internal tool, for finite-order Hecke characters over $\mathbb{Q}(\sqrt{-3})$. We record its consequences for the Lindelöf μ-function of ζ: $\mu(\sigma)=0$ for $\sigma \ge 7/8$, previously known only for $\sigma \ge 1$, and by convexity $\mu(\sigma) \le \frac{26}{63}(\frac{7}{8}-\sigma)$ on $[\frac{1}{2},\frac{7}{8}]$, which improves the sharpest known bounds for $\sigma > 1351/1976 \approx 0.684$; for instance $\mu(4/5) \le 13/420 = 0.0310$ against the previous 0.0415. Analogous interpolations hold in the conductor aspect. We also note that the standard single-abscissa theorems converting growth into zero-density bounds give nothing below 7/8 from this input.

1 Input and statement

Throughout, $\Theta := 7/8$. The input is:

(H) Every Dirichlet L-function $L(s,\chi)$, including ζ, has no zeros with $\operatorname{Re} s > \Theta$.

(H) is Theorem 1.1 of [12] (the Lean-verified statements there are exactly the nonvanishing of ζ and of all Dirichlet L-functions for $\operatorname{Re} s > 7/8$ [13]); by [9] one may take $\Theta = 0.874957\ldots$ and by [10] the same holds for finite-order Hecke L-functions over every number field. Everything below is conditional on (H) and is stated for $\Theta = 7/8$; the proofs work verbatim for any $\Theta \in (1/2,1)$.

For a primitive character χ modulo q put $Q := q(|t|+3)$ and, for the principal character ($q=1$), $L(s,\chi) := (s-1)\zeta(s)/(s+1)$ with its removable value at 1.

Theorem 1. *Assume (H). For every $\delta > 0$ and $\varepsilon > 0$, uniformly in all primitive χ modulo q, all real t and all $\sigma \ge \Theta+\delta$,*

$$|L(\sigma+it,\chi)| + |L(\sigma+it,\chi)|^{-1} \ll_{\delta,\varepsilon} Q^{\varepsilon}, \qquad \left|\frac{L'}{L}(\sigma+it,\chi)\right| \ll_{\delta} \log Q.$$

For $q=1$ the reciprocal bound passes to $1/\zeta$, and for $|t| \ge 1$ the upper bound passes to ζ.

This is [12, Lemma 4.9, "global assertion"] for finite-order Hecke characters over $\mathbb{Q}(\sqrt{-3})$; we give the (identical) proof for Dirichlet L-functions so that the statement is self-contained.

*Prepared with the assistance of Claude (Anthropic); the author takes responsibility for the content.

Proof. We may assume $\delta \le 1/8$: if $\delta > 1/8$ then $\sigma \ge \Theta+\delta > 1+(\delta-1/8)$ and the Euler product gives $|\log L(s,\chi)| \le \log\zeta(1+\delta-1/8)$ (for $q=1$ add the bounded logarithm of $(s-1)/(s+1)$), which implies all three bounds. Fix t and put $s_0 := 2+it$. Let g be the holomorphic branch of $\log L(s,\chi)$ on the half-plane $\{\operatorname{Re} s > \Theta\}$, which is simply connected and zero-free by (H) (for $q=1$ the regularized function has neither zero nor pole there), agreeing with the Euler-product logarithm for $\operatorname{Re} s > 1$ (for $q=1$ add the principal branch of $\log\frac{s-1}{s+1}$). Then g is holomorphic on a neighbourhood of every closed disk $\{|s-s_0| \le r\}$ with $r < 2-\Theta$. Put

$$r_0 := \tfrac{1}{2} < r_3 := 2-\Theta-2\delta < r_2 := 2-\Theta-\delta < r_1 := 2-\Theta-\delta/2,$$

so that all these closed disks lie in $\operatorname{Re} s \ge \Theta+\delta/2$. At the centre, $|g(s_0)| \le \log\zeta(2)+O(1)$. On the disk of radius r_1 one has $|L(s,\chi)| \le C_0 Q^A$ with absolute C_0, A: for $\operatorname{Re} s \ge 5/4$ by the Euler product, and for $-1/4 \le \operatorname{Re} s \le 5/4$ by Phragmén–Lindelöf between those two lines using the functional equation and Stirling's formula on $\operatorname{Re} s = -1/4$ (one may take $A = 3/4$; see e.g. [6, Ch. 10]); for $q=1$ apply this to $(s-1)\zeta(s)/(s+1)$, which is holomorphic in that strip. Points of the disk have height t' with $|t'-t| < 9/8$, so $q(|t'|+3) \le 2Q$, absorbed into C_0. Hence $\operatorname{Re} g = \log|L| \le A\log Q + \log C_0$ on $|s-s_0| \le r_1$, and the Borel–Carathéodory inequality (in the form of [7, §5.5]) on the radii $r_1 > r_2$ gives

$$\max_{|s-s_0|\le r_2} |g(s)| \;\le\; \frac{2r_2}{r_1-r_2}\,(A\log Q + \log C_0) + \frac{r_1+r_2}{r_1-r_2}|g(s_0)| \;\ll_\delta\; \log Q,$$

since $r_1 - r_2 = \delta/2$. On the disk of radius r_0 one has $\operatorname{Re} s \ge 3/2$, so $|g| \le \log\zeta(3/2)+O(1)$ there. Hadamard's three-circles theorem on the radii $r_0 < r_3 < r_2$ gives

$$\max_{|s-s_0|\le r_3} |g(s)| \;\ll_\delta\; (\log Q)^\theta, \qquad \theta = \frac{\log(r_3/r_0)}{\log(r_2/r_0)} < 1.$$

The disk of radius r_3 contains the segment $\{\sigma+it : \Theta+2\delta \le \sigma \le 2\}$ (its points are at distance $2-\sigma \le r_3$ from s_0); for $\sigma \ge 2$ the Euler product gives $|g| \le \log\zeta(2)+O(1)$. Since $\exp(C(\log Q)^\theta) \ll_{\theta,\varepsilon} Q^\varepsilon$, applying this to $e^g = L$ and $e^{-g} = 1/L$ proves the first bound with 2δ in place of δ. Cauchy's estimate for $g' = L'/L$ between the radii r_3 and r_2 (distance δ) gives the second. Replacing δ by $\delta/2$ throughout gives the statement; all constants depend only on $(\Theta,\delta,\varepsilon)$. □

Uniformity in q of the half-plane enters exactly once: the zero-freeness of the disks of radius $< 2-\Theta$ around $2+it$ for every χ, with the same Θ. All radii and θ, hence all constants, depend only on $(\Theta,\delta,\varepsilon)$. If one only had $L(s,\chi) \ne 0$ for $\operatorname{Re} s > \Theta_q$ with $\Theta_q \to 1$, the constants would depend on q and nothing uniform would follow. For $q=1$ and $|t| \ge 1$ the upper bound passes from the regularized function to ζ itself, since $|(s+1)/(s-1)| \le (|t|+3)/|t| \le 4$; the L'/L bound for $q=1$ is for the regularized function.

2 Consequences for the growth of ζ

Let $\mu(\sigma) := \limsup_{|t|\to\infty} \log|\zeta(\sigma+it)|/\log|t|$ be the Lindelöf function; μ is convex and non-increasing, $\mu(\sigma) = 0$ for $\sigma \ge 1$, and $\mu(1/2) \le 13/84$ (Bourgain [3]). Unconditionally $\mu(\sigma) = 0$ was known only for $\sigma \ge 1$; the sharpest known bounds on $[1/2, 1)$ are the exponent-pair bounds compiled in [1, Table 6.2] (Trudgian–Yang [15] and Heath-Brown), all strictly positive, e.g. $\mu(\sigma) \le \frac{29}{130} - \frac{3}{13}\sigma$ on $[0.8558, 0.9233]$, giving $\mu(7/8) \le 0.0212$.

Corollary 2. *Assume (H). Then $\mu(\sigma) = 0$ for $\sigma \geq 7/8$, and*

$$\mu(\sigma) \;\leq\; \frac{26}{63}\Big(\frac{7}{8} - \sigma\Big) \qquad (\tfrac{1}{2} \leq \sigma \leq \tfrac{7}{8}).$$

The right-hand side is smaller than the bound of [1, Table 6.2] exactly for $\sigma > 1351/1976 = 0.6837\ldots$ (the crossing is with the piece $\frac{1333}{3825} - \frac{1508}{3825}\sigma$). Sample values: $\mu(0.7) \leq 13/180 = 0.0722$ (previously 0.0726), $\mu(3/4) \leq 13/252 = 0.0516$ (0.0564), $\mu(4/5) \leq 13/420 = 0.0310$ (0.0415), $\mu(0.85) \leq 13/1260 = 0.0103$ (0.0272).

Proof. Theorem 1 with $q = 1$ gives $\zeta(\sigma + it) \ll_{\delta,\varepsilon} |t|^{\varepsilon}$ for $\sigma \geq 7/8 + \delta$, i.e. $\mu(\sigma) = 0$ for $\sigma > 7/8$ (using that the bound passes from the regularized function to ζ); μ is convex [1, Lemma 6.3] and finite on $\mathbb{R}$, hence continuous, so $\mu(7/8) = \lim_{\sigma \downarrow 7/8} \mu(\sigma) = 0$. Convexity between $\sigma = 1/2$ and $\sigma = 7/8$ gives $\mu(\sigma) \leq \mu(1/2)\,\frac{7/8-\sigma}{3/8} \leq \frac{13}{84} \cdot \frac{8}{3}\big(\frac{7}{8} - \sigma\big)$. The comparison with the table is a finite computation (script `lab/growth78/compare_mu.py`). □

Corollary 3 (conductor aspect)**.** *Assume (H). Let $c \geq 0$ and suppose that for a class $\mathcal{C}$ of primitive nonprincipal characters one has $L(\frac{1}{2} + it, \chi) \ll_{\varepsilon} (q(|t|+2))^{c+\varepsilon}$ for all $\chi \in \mathcal{C}$ and all t. Then for $\chi \in \mathcal{C}$, uniformly in t,*

$$L(\sigma + it, \chi) \;\ll_{\varepsilon}\; (q(|t|+2))^{\,c\,(7/8-\sigma)/(3/8)+\varepsilon} \qquad (\tfrac{1}{2} \leq \sigma \leq \tfrac{7}{8}).$$

In particular, with the convexity bound $c = 1/4$ (all q), the exponent $\frac{2}{3}(\frac{7}{8} - \sigma)$; with the Weyl-strength bound $c = 1/6$ of Petrow–Young [14] (all q), the exponent $\frac{4}{9}(\frac{7}{8} - \sigma)$, e.g. $q^{1/18+\varepsilon}$ at $\sigma = 3/4$ against the convexity exponent $1/8$ and the explicit Burgess-type q-aspect exponent $31/80 - 2\sigma/5 = 7/80$ of [4, Thm. 1.3].

Proof. Phragmén–Lindelöf for the entire function $L(s, \chi)$, of finite order in the strip by the functional equation, in $\frac{1}{2} \leq \operatorname{Re} s \leq \frac{7}{8} + \delta$, with the hypothesis on the left line and Theorem 1 on the right line; choose $\delta = \delta(\varepsilon)$ so that the exponent changes by at most ε (the constants of Theorem 1 depend on δ, hence on ε). □

Remark 4 (zero density)**.** *The standard theorems that convert a growth bound at one abscissa α into a zero-density bound (Montgomery [11, Thm. 12.3], in the form $A(\sigma) \leq \mu(\alpha) \cdot 2(3\sigma - 1 - 2\alpha)/((2\sigma - 1 - \alpha)(\sigma - \alpha))$ for $(\alpha + 1)/2 \leq \sigma \leq 1$, [1, Lemma 11.27]; the Halász–Montgomery inequality in the form of [1, Cor. 8.13]) only produce bounds for $\sigma \geq (1+\alpha)/2$. With $\alpha \geq 7/8$ this is $\sigma \geq 15/16$, where (H) already gives $N(\sigma, T) = 0$. Feeding the improved $\mu(\alpha)$ of Corollary 2 for $\alpha \in (0.684, 7/8)$ into Montgomery's theorem produces bounds for $\sigma \geq (1+\alpha)/2 > 0.842$; a direct computation (`lab/growth78/density_check.py`) shows that on $[0.842, 7/8)$ the resulting bound, minimized over α with $\mu(\alpha)$ replaced by the minimum of the table value and $\frac{26}{63}(\frac{7}{8} - \alpha)$, is attained as $\alpha \to 1/2$ and exceeds 2 (e.g. 2.06 at $\sigma = 7/8$, 2.43 at $\sigma = 0.85$), whereas [1, Table 11.1] gives $A(\sigma) < 2$ there; and [1, Cor. 8.13] with $\sigma' > 0.684$ yields Montgomery's large-values conjecture only for $\tau < 1$, where it is trivial. Ivić's large-values theorem [1, Thm. 8.15] is [1, Cor. 8.13] with the abscissa σ' optimised and μ replaced by a piecewise majorant; recomputing the Montgomery range $\sup_{\sigma' < 2\sigma - 1}(2\sigma - 1 - \sigma')/\mu(\sigma')$ with μ replaced by the minimum of the table value and $\frac{26}{63}(\frac{7}{8} - \sigma')$ (`lab/growth78/cor813_check.py`) shows that the supremum is attained at $\sigma' = 1/2$ for every $\sigma \leq 7/8$, so the improved μ on $(0.684, 7/8)$ never enters. We also checked, with the derivation engine of [1] (commit `c8eda5e`), that the zeta-large-value term of its Corollary 11.7 on $[0.7, 7/8]$ is unchanged by the new μ bounds for $\tau_0 \in \{3, 5\}$. So nothing new below $7/8$ follows from (H) by any of the μ-based routes: the growth improvement of Corollary 2 is real but does not propagate to zero density.*

Remark 5. *Nothing here is new as a method; the content is the explicit recording of what (H) gives, which none of [12, 9, 10, 16, 2, 8] states. If (H) holds as asserted in [12, 13], this is the first instance of $\mu(\sigma) = 0$ at an abscissa $\sigma < 1$; we have not independently checked [12] and state everything conditionally on (H).*

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